TALENTRA DATABASE DESIGN HACKATHON

Milestone 6 — Keys and Constraints Workbook (B20–B23)

Status: Working implementation guide. It is not a validated schema and does not constitute team sign-off.

1. What Milestone 6 requires

The docket defines B20 as adding keys and constraints to Profile, B21 as validating them, B22 as adding keys and constraints to Jobs and Applications, and B23 as validating them. B23 produces the signed Phase 1 design. Start this work only after the relevant Milestone 5 design is validated.

2. Standards to apply to every in-scope entity

Area	Required rule
Names	Singular snake_case table names; PK column named id; FK name = referenced table + _id; indexes named idx_table_column; unique constraints named uq_table_column.
Primary keys	Every entity has a surrogate PK. Use one consistent identifier type; BIGINT identity is a proposal from the working design package, not a docket-mandated type.
Foreign keys	Enforce every FK in the database and document ON UPDATE and ON DELETE. Prefer RESTRICT for retained history; use CASCADE only for dependent children when appropriate.
Unique rules	Natural keys/reference codes are UNIQUE, not PKs. Add business uniqueness only after the team confirms the rule (e.g. normalized skill code, candidate/job application uniqueness).
CHECK rules	Use checks for end_date >= start_date, salary minimum <= maximum when both supplied, positive vacancy count, nonnegative money, and bounded scores where relevant.
Lookup domains	Fixed enumerations use lookup/reference tables with code, localized display name, active flag, and sort order; do not store arbitrary free-text statuses.
Audit / lifecycle	Business tables include created_at, created_by, updated_at, updated_by. State soft/hard-delete policy, deleted_at when soft deletion applies, retention rules, and append-only history/audit behaviour.
Data standards	UTC timestamps (PostgreSQL TIMESTAMPTZ if PostgreSQL remains selected); fixed-precision NUMERIC plus currency code for money; documented nullable rationale; Public/Internal/Personal/Sensitive classification.

3. B20 — Profile keys and constraints

Apply these checks to the approved Profile entity list and its column dictionary. Do not add tables solely because they appear in this example; match the team's validated inventory.

Candidate area	Key / constraint checklist
candidate_profile	PK id; FK to account/user dependency as agreed; decide whether user_id is UNIQUE (one profile per account) or non-unique (multiple profiles). Add name-length rules and required audit fields.
candidate_education	PK id; FK candidate_profile_id; FK qualification_id to Master Data; CHECK end_date IS NULL OR start_date IS NULL OR end_date >= start_date.
candidate_experience / candidate_work_experience	Choose one canonical entity name. PK id; FK candidate_profile_id; CHECK end_date IS NULL OR start_date IS NULL OR end_date >= start_date; document current-job representation.
candidate_skill / profile skill mapping	PK id or composite mapping only if the approved design permits; normalized skill FK; add UNIQUE(candidate_profile_id, skill_id) if duplicate profile-skill rows are not allowed.
candidate_language / profile language mapping	Use language lookup FK; if proficiency is stored, use lookup or CHECK-bounded value. Enforce one row per profile/language if duplicates are disallowed.
resume_version / project / assessment / course records	PK id; FKs to profile and relevant owning module; preserve version/history if required; validate scores/ranges and date windows; state file storage vs external reference.
Profile history tables	Append-only where required; index common lookup FKs and timestamps; do not cascade-delete retained audit/history rows without explicit policy.

4. B21 — Validate Profile keys and constraints

• Every approved Profile entity has exactly one declared surrogate PK and a consistent identifier type.

• Every FK target exists in its owning module or is explicitly marked as an external dependency; ON UPDATE and ON DELETE are recorded.

• Natural identifiers and approved business uniqueness rules have named UNIQUE constraints.

• CHECK constraints cover applicable date ranges, scores, nonnegative amounts, and other expressible rules.

• Audit fields, deletion/retention policy, nullable justifications, and data classifications are complete.

• The ERD, column dictionary, DDL, and validation results use identical entity and column names.

• Record each check as PASS / FAIL / N/A with evidence and reviewer; do not mark PASS without evidence.

5. B22 — Jobs & Applications keys and constraints

Entity / area	Key / constraint checklist
job_requirement	PK id; FK to organisation dependency and Master Data role/type/status lookups; CHECK minimum_salary >= 0, maximum_salary >= 0, and minimum_salary <= maximum_salary when both supplied; require currency_code when a salary is present.
job_position	PK id; FK job_requirement_id; CHECK vacancy_count > 0; location FK to owning module; define whether positions sum to requirement headcount.
job_questionnaire / questionnaire_question	PK id on each; FK questionnaire to job_requirement and question to questionnaire; enforce order/number rules only if agreed. If answers are in scope, agree response table and one-vs-many answer cardinality first.
application	PK id; FK to candidate_profile and job_requirement; add business UNIQUE constraint only after repeat/withdraw/reapply policy is decided. Index FKs and frequently filtered status/date columns.
application_status_history	If approved in the design: PK id; FK application_id; append-only; status FK to lookup; actor and event timestamp required; avoid contradictory current status.
shortlist / interview / offer / joining / rejection	Use approved canonical entity names. Enforce FKs to the owning application/job and actor dependencies; define offer cardinality, interview scheduling constraints, and outcome/status consistency.
search and application history	Confirm whether history is event/audit data or a derived view. Preserve required actor, subject, timestamp, and outcome fields; index according to query patterns.

6. B23 — Validate Jobs & Applications and sign Phase 1

• All keys and constraints match the approved entity, relationship, and column inventories.

• No FK points to a nonexistent or inconsistently named table/column.

• Each FK has explicit ON UPDATE and ON DELETE behaviour; retained history is protected from accidental cascade deletion.

• All approved natural/business uniqueness rules and CHECK constraints are present and tested with both valid and invalid examples.

• Database-level concurrency/licence/session rules are documented where applicable; application-only checks are not treated as sufficient.

• Index choices are justified by expected queries and benchmark workloads; avoid redundant indexes that duplicate UNIQUE/FK indexes.

• DDL applies cleanly to a fresh database; constraint tests and representative inserts/updates/deletes pass.

• Record reviewer names, date, test evidence, unresolved risks, and the team's explicit sign-off for the Phase 1 design.

7. Suggested SQL patterns

Examples only — adapt table/column names to the approved inventory and the selected database engine.

```
-- Date range
CONSTRAINT chk_candidate_education_dates
CHECK (end_date IS NULL OR start_date IS NULL OR end_date >= start_date)

-- Salary range
CONSTRAINT chk_job_requirement_salary_range
CHECK (minimum_salary IS NULL OR maximum_salary IS NULL
OR minimum_salary <= maximum_salary)
```

```
-- Positive vacancy count
CONSTRAINT chk_job_position_vacancy_count
CHECK (vacancy_count > 0)

-- Example FK with explicit lifecycle
CONSTRAINT fk_application_job_requirement
FOREIGN KEY (job_requirement_id)
REFERENCES job_requirement(id)
ON UPDATE RESTRICT ON DELETE RESTRICT

-- Example uniqueness, only if team approves this rule
CONSTRAINT uq_candidate_skill_profile_skill
UNIQUE (candidate_profile_id, skill_id)
```

8. Final status note

This workbook is a proposed checklist and design aid. It does not prove B20–B23 are complete. The team must validate against its approved entity list, run DDL/constraint tests, record evidence, and sign the Phase 1 design before treating Milestone 6 as done.